G6PD Deficiency

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Etiology

- ☑ *Glucose-6-phosphate dehydrogenase (G6PD) deficiency* is a hereditary condition in which red blood cells breakdown (hemolysis) when the body is exposed to certain foods, drugs, infections, or stress.
- ☑ Glucose-6-phosphate dehydrogenase deficiency is caused by mutations in *G6PD* gene leading to deficiency in G6PD enzyme and oxidative damage to red blood cells.

Inheritance

Glucose-6-phosphate dehydrogenase is inherited in an X-linked pattern. Males are affected more than females. This is because males have only one copy of the *G6PD* gene.

Prevalence

- ☑ Glucose-6-phosphate dehydrogenase mutation may be partially protective against malaria. Hence, it is common in malaria belt countries. About 400 million people worldwide have this deficiency. It is more seen in certain areas of Africa, Asia, Mediterranean, and Middle East.
- ☑ It varies from 0 to 27% in different caste, ethnic, and linguistic groups. The frequency is higher among the tribals. Before starting antimalarial treatment, especially in tribal males, G6PD activity should be assessed.

Signs and Symptoms

- ☑ It may be seen in neonatal period—mild-to-severe jaundice requiring therapy.
- ☑ Other patients—sudden onset of pallor, jaundice, dark urine, fatigue, shortness of breath, and tachycardia when triggered.
- ☑ Triggers can be bacterial, viral infections, certain medicines including antibiotics and anti-malarials, exposure to fava beans, naphthalene balls, etc.
- ☑ Many people with this condition never experience symptoms.

Diagnosis

- ☑ Can be diagnosed in newborn period by heel stick newborn blood sampling.
- ☑ During acute hemolysis, severe anemia, high reticulocyte counts, and indirect hyper-bilirubinemia.
- ☑ *Complete blood count* (CBC), reticulocyte count, and good peripheral smear evaluation must have:
  - Anemia, fall in hemoglobin (Hb) by 3–4 g/dL
  - Anemia is normocytic normo-chromic and neutrophilic response may be seen.
  - Marked anisopoikilocytosis with raised red blood cell distribution width (RDW)
  - There is presence of micro-spherocytes, "bite" cells, "blister cells," and Heinz bodies.

Heinz bodies are small inclusion bodies (denatured Hb) typically seen in the early phase of hemolysis demonstrated by supravital staining. These are pinched off by the spleen leaving the bite cells in circulation **(Fig. 1)**.

- ☑ *Serum haptoglobin* is reduced and urine Hb is positive
- ☑ G6PD enzyme activity—*Enzyme assays.*

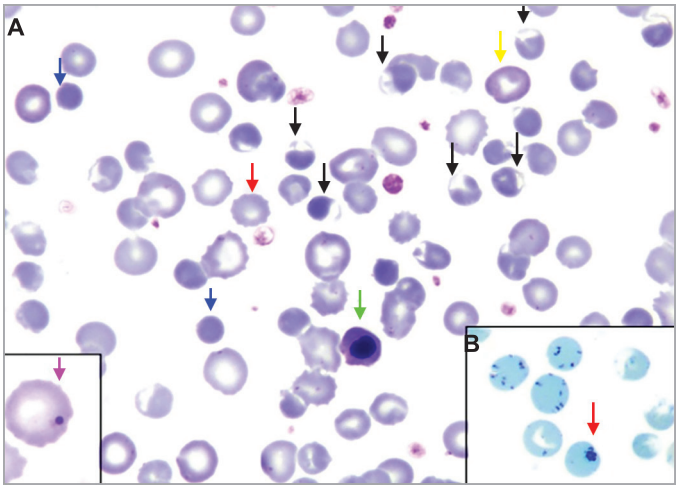

Fig. 1: Peripheral blood smear (PBS)—numerous blister cells (black arrow), spherocytes (blue arrow), polychromasia (yellow arrows), nucleated red blood cell (green arrow), Howell–Jolly body (**A**, pink arrow), reticulocyte preparation revealed Heinz bodies (**B**, orange arrow), and acanthocytes (red arrow) signifies acute renal injury.

Diagnosis

Sample

Ideal—dipotassium ethylenediaminetetraacetic acid (K2-EDTA); however, citrate or lithium heparin could be used. G6PD activity is stable at 4–8°C up to 72 hours. All samples should be transported in heat-insulated envelopes and refrigerated transport containers.

Time of Testing–Avoid At

- ☑ *Acute hemolysis*: Young erythrocytes and reticulocytes have up to five times higher G6PD enzyme activity and can give false negative test.
- ☑ *Post blood transfusion*: False negatives also may occur after blood transfusion, and retesting may be required after 2–3 months when erythrocytes of all ages are present.

Types of Assays

- ☑ Semiquantitative assay
- ☑ *Quantitative assay*: Spectrophotometric assay is the gold standard in diagnosing the deficiency. Normal activity is considered to be 7–10 IU/g Hb when measured at 30°C
- ☑ Flow cytometric evaluation. Point-of-care rapid diagnostic tests.

Mutation Analysis

Molecular analysis holds importance in the diagnosis of heterozygous females with intermediate enzyme activity (due to lyonization).

Types

- ☑ The G6PD A—this is the most described variant—(202 G > A and 376A > G) which is common in Africa and persons of African descent (especially in the Americas).
- ☑ G6PD* Mediterranean—the mutation (563C > T) is common in Southern Europe and the Middle East—related to favism.
- ☑ G6PD* Mahidol, Viangchan, and Other Asian variants
- ☑ Other variants—San Diego, Chatham, Canton, Verona, etc.

Classes

Different classes of G6PD have been depicted in **Table 1**.

TABLE 1: Different classes of glucose-6-phosphate dehydrogenase (G6PD).

Class	*G6PD activity*	*Hemolysis*	*Most common variants*
I	<10%	Chronic	G6PD San Diego
II	<10%	Acute	G6PD Mediterranean, G6PD Chatham
III	10–60%	Acute	G6PD A–, G6PD Canton
IV	60–150%	Absent	G6PD B, G6PD A+
V	Increased	G6PD Verona	G6PD Verona

Management and Prevention

- ☑ Discontinue the offending oxidative stressor, most commonly a medication.
- ☑ Most patients recover spontaneously and require monitoring.
- ☑ If the anemia very severe (preexisting anemia, or if the rapid decline in Hb), then a packed cell transfusion can be lifesaving.
- ☑ If Hb <7 g/dL, consider *packed cell* transfusion.
- ☑ If there are no signs of hemoglobinuria, packed cell tranfusion not required, rather the patient should be monitored closely for 48 hours.
- ☑ Consideration of comorbidities, hemodynamic stability, and the patient's previous hemolytic experience to decide.

- ☑ Oxygen and fluids is critical during acute crisis.
- ☑ Acute kidney injury may be due to hypovolemic shock or hemoglobinuria. Renal function usually recovers with time.
- ☑ Patient education about avoiding the triggers, diet, and lifestyle modifications and recognition of signs and symptoms helps to prevent further exacerbations.
- ☑ Splenectomy—usually not indicated, except in patients where hypersplenism is severe enough to increase dependency on transfusions.
- ☑ Folic acid and iron supplements may be useful although G6PD deficiency usually is asymptomatic and the associated hemolysis usually is short-lived.
- ☑ Phototherapy for neonatal jaundice is dependent on bilirubin levels.

TABLE 2: Definite risk of hemolysis.

Pharmacological class	***Drugs***
Anthelminitics	☑ β-Naphthol ☑ Niridazole ☑ Stibophen
Antibiotic	☑ Nitrofurans – Nitrofurantoin – Nitrofurazone ☑ Quinolones – Ciprofloxacin – Moxifloxacin – Nalidixic acid – Norfloxacin – Olfoxacin ☑ Chloramphenicol ☑ Sulfonamides – Co-trimoxazole (Sulfamethoxazole + Trimethoprim) – Sulfacetamide – Sulfadiazine – Sulfadimidine – Sulfamethoxazole – Sulfanilamide – Sulfapyridine – Sulfasalzine (Salazosulfapyridine) – Sulfisoxazole (Sulfafurazole)

Contd...

Contd...

Pharmacological class	***Drugs***
Antimalarials	☑ Mepacrine ☑ Parmaquine ☑ Pentaquine ☑ Primaquine
Antimethemoglobinemic agents	Methylene blue
Antimycobacterials	☑ Dapsone ☑ Para-aminosalicylic acid ☑ Sulfones – Aldesulfone sodium (Solfozone) – Glucosulfone – Thiazosulfone
Antineoplastic adjuncts	☑ Doxorubicin ☑ Rasburicase
Genitourinary analgesics	Phenazopyridine (Pyridium)
Other	☑ Acetylphenylhydrazone ☑ Phenylhydrazine

TABLE 3: Possible risk of hemolysis.

Pharmacological class	***Drugs***
Analgesics	☑ Acetylsalicylic acid (Aspirin) ☑ Acetanilide ☑ Paracetamol (Acetaminophen) ☑ Aminophenazone (Aminopyrine) ☑ Dipyrone (Metamizole) ☑ Phenacetin ☑ Phenazone (Antipyrine) ☑ Phenylbutazone ☑ Tiaprofenic acid
Antibiotic	☑ Furazolidone ☑ Streptomycin ☑ Sulfonamides – Sulfacytine – Sulfaguanidine – Sulfamerazine – Sulfamethoxypyridazone

Contd...

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Pharmacological class	***Drugs***
Anticonvulsants	☑ Phenytoin
Antidiabetic	☑ Glibenclamide
Antidotes	☑ Dimercaprol (BAL)
Antihistamines	☑ Antazoline (Antistine) ☑ Diphenhydramine ☑ Tripelennamine
Antihypertensives	☑ Hydralazine ☑ Methyldopa
Antimalarials	☑ Chloroquine and derivatives ☑ Proguanil ☑ Pyrimethamine ☑ Quinidine ☑ Quinine
Antimycobacterials	☑ Isoniazid
Antiparkinsonism agents	☑ Trihexyphenidyl (Benzhexol)
Cardiovascular drugs	☑ Dopamine (L-dopa) ☑ Procainamide ☑ Quinidine
Diagnostic agent for cancer	☑ Detection ☑ Tuluidine blue
Gout preparations	☑ Colchinine ☑ Probenecid
Hormonal contraceptives	☑ Mestranol
Nitrates	☑ Isobutyl nitrite
Vitamin K substance	☑ Menadiol Na sulfate ☑ Menadione ☑ Menadione Na bisulfite ☑ Phytomenadione
Vitamins	☑ Ascorbic acid (Vitamin C) (rare)
Other	☑ Arsine ☑ Berberine (in Coptis chinensis) ☑ Fava beans ☑ Naphthalene (in Mothballs) ☑ Para-aminobenzoic acid

Further Reading

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Megaloblastic Anemia

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